ANSWER KEY & MARKING SCHEME SET A

Chapter 6: **The Making of a Scientist** (Footprints Without Feet) | Class 10 English | Maximum Marks: 20

SECTION — A — Answers — 6 Marks

Q.	Expected Answer / Value Points	Marks
1	**(a)** Richard Ebright — *Winner of the Searle Scholar Award*	1
2	**(b)** how cells work — *Published in a scientific journal*	1
3	**(a)** the journal had never before published the work of college students — *Proceedings of the National Academy of Science*	1
4	**(d)** was an only child with no team to play with — *He could not play team sports alone*	1
5	**(b)** rocks, fossils and coins — *Also he was an eager astronomer*	1
6	**(d)** encouraged him by buying equipment and giving company — *She bought telescopes and microscopes*	1

SECTION — B — Answers — 4 Marks

Q.	Expected Answer / Value Points	Marks
7	**(i)** He was an only child and could not play team sports.	1
	(ii) Butterflies and rocks (or fossils, coins).	1
	(iii) She bought equipment and spent evenings with him.	1
	(iv) She was supportive and devoted.	1

SECTION — C — Answers — 4 Marks

Q.	Expected Answer / Value Points	Marks
8	As an only child without playmates he turned to collecting butterflies, rocks, fossils and coins.	2
9	'The Travels of Monarch X' invited readers to tag butterflies for research and led him from collecting to real science.	2

SECTION — D — Answers — 6 Marks

Q.	Expected Answer / Value Points	Marks
10	An only child, he collected butterflies, rocks, fossils, coins and watched stars; his mother bought telescopes, microscopes and cameras, spent evenings helping him and encouraged him to learn.	3
	OR: Through the book 'The Travels of Monarch X', tagging butterflies for Dr Urquhart, raising monarchs in the basement and doing experiments for science fairs.	3
11	Seventh grade: slides, lost; eighth grade: viral disease in caterpillars, won; ninth: viceroy copying monarch; later: gold spots and hormone research, culture of wing cells.	3
	OR: Curiosity, perseverance, hard work, willingness to learn from failure, supportive family and teachers, competitive spirit and discipline.	3

Evaluator's note: Answers given are model answers / value points; accept any other relevant and correct response in the candidate's own words. For writing tasks, marks are for format, content, organisation and accuracy of language. In internal-choice questions, evaluate only the option attempted.

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Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapter 6: **The Making of a Scientist** (Footprints Without Feet) | Class 10 English | Maximum Marks: 20

SECTION — A — Answers — 6 Marks

Q.	Expected Answer / Value Points	Marks
1	**(d)** third grade — *His mother said Richie was her whole life*	1
2	**(d)** all twenty-five butterfly species of his hometown — *He was very determined*	1
3	**(c)** The Travels of Monarch X — *It told of monarch migration*	1
4	**(a)** butterfly migrations — *Ebright tagged monarchs for him*	1
5	**(c)** It was tedious with little feedback — *Only two of his tagged butterflies were recaptured*	1
6	**(d)** only displayed slides without doing a real experiment — *He learnt a lesson*	1

SECTION — B — Answers — 4 Marks

Q.	Expected Answer / Value Points	Marks
7	**(i)** Slides of frog tissues under a microscope.	1
	(ii) He had not done a real experiment.	1
	(iii) To do a real experiment the next year.	1
	(iv) Dr Urquhart.	1

SECTION — C — Answers — 4 Marks

Q.	Expected Answer / Value Points	Marks
8	That real science needs experiments, not just a neat display; he decided to do real experiments.	2
9	They produce a hormone necessary for the butterfly's full development.	2

SECTION — D — Answers — 6 Marks

Q.	Expected Answer / Value Points	Marks
10	He found a hormone from the gold spots; his X-ray photos of hormone structure led to a theory on how a cell reads DNA; he and Wong wrote the paper; it could help prevent diseases such as cancer.	3
	OR: He directed the tagging programme, sent ideas for experiments, and supported the research on gold spots.	3
11	A straight-A student, active in debate and Model United Nations, graduated from Harvard with highest honours.	3
	OR: It traces how experiences, hobbies, curiosity and encouragement led Ebright to become a scientist.	3

Evaluator's note: Answers given are model answers / value points; accept any other relevant and correct response in the candidate's own words. For writing tasks, marks are for format, content, organisation and accuracy of language. In internal-choice questions, evaluate only the option attempted.